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Unit – 5 : Session – 2 : SLO – 1 | Graph Representation — Adjacency Matrix

■ I. Fill in the Blanks

- In an adjacency matrix for a graph with n vertices, the size of the matrix is $n \times n$.
- If there is an edge between vertex i and vertex j , then $\text{adjacency_matrix}[i][j] = 1$.
- In an undirected graph, the adjacency matrix is always SYMMETRIC.
- In a directed graph, $\text{adjacency_matrix}[i][j] = 1$ means there is a directed edge from i to j .
- The space complexity of an adjacency matrix representation is $O(n^2)$.

■ II. True or False

- The adjacency matrix is suitable for representing sparse graphs — FALSE (it wastes space; lists suit sparse graphs).
- In a weighted graph, adjacency matrix elements can store the weight values instead of 1s and 0s — TRUE.
- In an undirected graph, $\text{adjacency_matrix}[i][j]$ will always equal $\text{adjacency_matrix}[j][i]$ — TRUE.
- Adjacency matrices use less memory compared to adjacency lists for dense graphs — TRUE.

■ III. Short Answers

- Adjacency matrix: a 2-D $n \times n$ array where entry $[i][j]$ is 1 (or the edge weight) if an edge exists between vertex i and vertex j , and 0 otherwise.
- Advantages: (1) edge existence checked in $O(1)$ time; (2) simple to implement and easy to store weights.

■ IV. Adjacency Matrix — Undirected Graph (edges A-B, A-C, B-D, C-D, B-E, D-E)

- A B C D E
- A [0 1 1 0 0]
- B [1 0 0 1 1]
- C [1 0 0 1 0]
- D [0 1 1 0 1]
- E [0 1 0 1 0]

■ V. Adjacency Matrix — Directed Graph (A->A, A->B, A->C, B->E, D->B, D->C, D->E)

- A B C D E
- A [1 1 1 0 0]
- B [0 0 0 0 1]
- C [0 0 0 0 0]
- D [0 1 1 0 1]
- E [0 0 0 0 0]

■ **VI. Adjacency Matrix — Undirected Weighted Graph (A-B=5, A-C=2, B-D=3, C-D=4, B-E=6, D-E=8)**

- A B C D E
- A [0 5 2 0 0]
- B [5 0 0 3 6]
- C [2 0 0 4 0]
- D [0 3 4 0 8]
- E [0 6 0 8 0]

■ **VII. Solve and Write — Given 4-Vertex Matrix**

- Edges from matrix: (1,2), (1,3), (2,4), (3,4).
- Graph: 1 connects to 2 and 3; 4 connects to 2 and 3 (a 4-cycle 1-2-4-3-1).
- Vertices adjacent to vertex 2: vertices 1 and 4.
- Number of edges: 4 (the matrix has 8 ones; each edge counted twice in an undirected graph).